

Sahil Suriya

AI / ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with 1+ year of production experience building end-to-end ML pipelines, RAG systems, and NLP-powered services. I specialize in Python, LangChain, FastAPI, and cloud deployment. Published researcher (2 peer-reviewed papers) and GHCI Hackathon 2024 participants. Passionate about shipping AI systems that deliver measurable business impact.

EXPERIENCE

AI/ML Engineer | Vector Stack, Bangalore

Sep 2025 – Present

- Developed and deployed **3 FastAPI microservices** powering core ML workflows, handling **1,000+ daily API calls** with 99.9% uptime.
- Designed a **video-transcript synchronization system** achieving **99.5% alignment accuracy** using sequence alignment algorithms.
- Optimized **50+ API endpoints**, reducing error rates by **20%** and improving response latency by ~35% through caching and async refactoring.
- Collaborated with the ML team to integrate model inference pipelines into production, cutting deployment cycle time by **40%**.

AI Data Curation and RLHF Intern | Skypoint Cloud, Bangalore

Mar 2024 – May 2024

- Automated web scraping workflows using **Python + Selenium**, improving data collection efficiency by **30%** (~2.5 hrs saved daily).
- Built preprocessing pipelines delivering **95% clean datasets**, directly reducing model retraining frequency.
- Engineered an **AI ingestion pipeline** that reduced model processing time by **25%**, saving ~2 hrs per training run.

PROJECTS

RAG-Based Multi-Format Document QA Chatbot

- Built an **agent-based RAG system** using **LangChain + FAISS**, enabling retrieval across **100+ document types** (PDF, DOCX, HTML, CSV).
- Designed custom **LLaMA-based retrievers** improving extraction accuracy from **65% to 85%** on internal benchmarks.
- Reduced analyst document-search time by **70%** in pilot testing with 3 internal teams.

AI Voice Assistant for Financial Transactions

- Engineered a secure AI voice assistant using **Gemini LLM + Google Speech-to-Text (STT)** enabling natural language financial transactions.
- Achieved **95%+ voice command recognition accuracy** across 10 transaction types in real-world conditions.
- Presented at GHCI Hackathon 2025, Bangalore** — selected among top projects at India's largest tech conference.

News Summarization & Sentiment Analysis

- Built a **FastAPI NLP service** aggregating and analyzing news across **15+ data sources** using transformer-based summarization models.
- Designed comparative sentiment scoring using NLP to identify competitive trends and market signals.
- Reduced manual analysis time by **90%**, delivering automated daily digests to 5 business stakeholders.

PUBLICATIONS

- Automated Timetable Generator System** — automated academic scheduling using constraint-based optimization. Peer-reviewed journal publication.

- **BLOG AI Project** — AI-powered long-form content generation using NLP and transformer fine-tuning. Peer-reviewed journal publication.

EDUCATION

Master of Computer Applications (MCA) | Dr. Ambedkar Institute of Technology, Bangalore

Dec 2023 – Sep 2025

Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Database Systems, Data Structures & Algorithms

Bachelor of Computer Applications (BCA) | BMS College of Commerce & Management, Bangalore

Aug 2020 – Aug 2023

TECHNICAL SKILLS

Languages: Python, SQL, R

ML / AI: Scikit-learn, TensorFlow, LangChain, FAISS, Chroma, HuggingFace, OpenCV, NLP, Deep Learning

Data Engineering: ETL Pipelines, Apache Airflow, PostgreSQL, PySpark, Pandas, NumPy

Deployment: FastAPI, Flask, REST APIs, AWS, Docker (basic), Git, GitHub

Visualization: Power BI, Tableau, Matplotlib, Seaborn

Core Concepts: Regression, Classification, Clustering, Retrieval-Augmented Generation (RAG), Model Evaluation

CERTIFICATIONS

- **Python for Data Science** — Indian Institute of Technology (IIT) Madras
- **Python for Data Science, AI & Development** — IBM
- **Machine Learning** — Simplilearn

ACHIEVEMENTS

- **GHCI Hackathon 2025, Bangalore** — Presented AI Voice Assistant; recognized among top projects at India's largest tech event.
- Published **2 peer-reviewed research papers** on AI-powered systems (scheduling optimization and content generation).

LANGUAGES

English (Professional) • Hindi (Native)